# Nazim Belabbaci

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## Summary

AI research scientist (PhD, Computer Science) and Postdoctoral Research Fellow at the Center for Virology and Vaccine Research (CVVR), BIDMC, a Harvard Medical School teaching hospital. I build AI for complex biomedical data end to end, from genomic sequences and multi-omics to wearable physiological signals, with a focus on models that hold up when data are scarce and noisy. Reviewer for Future Generation Computer Systems, Results in Engineering, and Digital Health.

## Research & Academic Experience

**Postdoctoral Research Fellow**, *Beth Israel Deaconess Medical Center (BIDMC)*, Boston, MA | 2026–Present

◊ Develop AI and multi-omics integration methods for an NIH-supported HIV research program, combining single-cell RNA, chromatin, and viral integration-site data to characterize host–virus interactions.

◊ Lead the AI modeling effort in a multidisciplinary influenza vaccine project, developing deep-learning models of antigenic evolution to guide vaccine strain selection.

**Graduate Research Assistant**, *University of Massachusetts Lowell*, Lowell, MA | 2022–2026

◊ Led the development of an AI-enabled optical wearable for hydration monitoring, including authoring the IRB protocol, collecting and curating physiological datasets, developing machine-learning models, deploying on-device inference pipelines, and validating system against saliva ground-truth measurements.
Formal invention disclosure filed with the UML's Office of Technology Commercialization.

◊ Developed a class-conditional latent diffusion model that generates synthetic FTIR spectroscopy data to address data scarcity and class imbalance in blood-based endometrial cancer detection. Augmenting training data with synthetic spectra improved accuracy and AUC by 4% and specificity by more than 12%, outperforming SMOTE and GAN-based balancing (PhD dissertation, 2026).

◊ Developed multimodal machine-learning models for fatigue, emotion, and activity recognition using physiological and motion signals (heart rate, respiration, EDA, temperature, accelerometry), achieving up to 15 percentage-point accuracy improvements over published baselines.

**PhD Student Researcher**, *University of Massachusetts Lowell*, Lowell, MA | Summer 2025

◊ Lead Researcher on an NSF-funded initiative developing end-to-end AI pipelines for real-time lead detection in water supplies, addressing critical public health and environmental safety risks through analytically validated (AAS-grounded) methodologies.

**PhD Research Intern**, *Beth Israel Deaconess Medical Center (BIDMC)*, Boston, MA | Summer 2025

◊Built a reproducible pipeline to analyze large RNA-sequencing datasets from multiple viruses (HIV, SARS-CoV-2, influenza, and mpox), automating statistical filtering and downstream functional analysis to identify virus-associated alternative splicing.

◊Extracted COVID symptoms from unstructured clinical notes using zero-shot classification (BART), then linked symptoms to transcriptomic pathway activity scores to quantify symptom–pathway relationships in COVID. Contributed to a paper published in *Nature Immunology*.

**Graduate Teaching Assistant**, *University of Massachusetts Lowell*, Lowell, MA | 2023–2026

◊ Teaching AI and Assembly Language; leading weekly laboratories and recitations for 100+ students.

**Graduate Research Assistant**, *University of North Dakota,* Grand Forks, ND | 2021–2022

◊ Invited to join NSF-funded research supporting AI-driven drug discovery, generating large-scale molecular simulation datasets using GROMACS on HPC systems, with abstract presented at the APS March Meeting 2022.

**Graduate Research Assistant**, *University of North Dakota,* Grand Forks, ND | Summer 2022

◊ Developed interpretable machine-learning models for non-invasive cancer detection using cell-free DNA, contributing to a peer-reviewed publication in Communications Biology.

◊ Co-developed MotifXplorer, a web-based platform for visualizing AI predictions and model interpretability, enabling researchers to link transcription-factor binding motifs to disease-relevant genomic outcomes.

## Education

**PhD**, Computer Science, University of Massachusetts Lowell, Lowell, MA | May 2026
**Master** Computer Science, University of Massachusetts Lowell, Lowell, MA | 2024
**Master** Computer Engineering, National Institute of E&E Engineering, Boumerdès, DZ | 2019
**Bachelor** Electrical & Electronic Engineering, National Institute of E&E Engineering, Boumerdès, DZ | 2017

## Industry Experience

**Co-Founder & Embedded Systems Engineer**, *Deadline Technologies*, Bouira, DZ | 2019–2021

◊ Designed Wi-Fi/BLE-enabled embedded systems for sensor data acquisition and telemetry, managing the full engineering lifecycle from PCB design and firmware development to deployment.

**Embedded Software Engineer**, *EURL Microtechnologies Lab*, Boumerdès, DZ | 2018–2019

◊ Developed firmware and software for laboratory-grade electrical measurement instruments used in national educational institutions, contributing to design specification, validation, and deployment.

## Awards & Honors

**Recipient, LaTorre Family Scholarship** Endowment Fund (UML) | 2023.
**Finalist** of the UND **Middleton School of Entrepreneurship Business Plan Competition** | 2022.

**Winner of the UND Big Idea Challenge** | 2022

◊ Awarded top honors for an AI-based early COVID-19 detection system, evaluated by industry experts through a comprehensive business and technical review.

**Winner of Leapfrog International Hackathon of Algiers** | 2018

◊Led a 48-hour team sprint that delivered a multi-sensor health monitoring prototype and real-time dashboard. Programmed a micro-controller to interface with custom circuitry as well as a mobile app.

## Civic Engagement

**Chair, Student Initiative**, *AAASTE*, Pembroke Pines, FL | 2022-2024

◊ Led a student initiative organizing science, technology, and entrepreneurship webinars, coordinating a team of 20 students to promote education and research engagement within the Algerian American science community.

**President & Head Project Manager**, *Wameedh Scientific Club*, Boumerdès, DZ | 2016-2019

◊ Led workshops and outreach programs, supporting 100+ students in developing technical skills. Secured funding, resources, and institutional partnerships to sustain and expand the club's scientific initiatives; winner of Algeria's National Scientific Club Contest (2018).